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Source apportionment of volatile organic compounds in a Philadelphia fenceline community: results from THRIVEair

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BACKGROUND: Urban exposure to volatile organic compounds (VOCs) is associated with several adverse health outcomes, including cancers and respiratory outcomes. We conducted a year-long VOC monitoring campaign in South Philadelphia to evaluate the impacts of traffic and redevelopment of a former oil refinery site on local air quality. Benzene was the community's primary pollutant of concern, but we measured a broader suite of VOCs to enable source apportionment, as presented here.

METHODS: In partnership with Philly Thrive, a Philadelphia-based environmental justice organization, we developed a community-responsive monitoring design and collected one-week integrated samples of 38 VOCs at 20 sites from July 2023 to June 2024 using passive thermal desorption tubes. We summarized concentrations of VOCs by land use and traffic density, and identified potential sources using toluene:benzene (T/B) ratio and (*m,p*)-xylenes:ethylbenzene (X/E) ratios. We employed positive matrix factorization (PMF) to apportion VOC sources.

RESULTS: VOC concentrations were generally higher in industrial vs. non-industrial and high- vs. low-traffic areas. Benzene, the compound of greatest community concern, occurred at low concentrations overall (median 1.21 µg/m³; range: 0.31–9.05 µg/m³), while (*m,p*)-xylenes emerged as a pollutant of concern in industrial areas. T/B and X/E ratios were higher among industrial and high-traffic sites, suggesting fresh (less photochemically aged) emissions of benzene derivatives. PMF identified five VOC sources: (1) vehicular exhaust, (2) gasoline evaporation, (3) industrial solvents, (4) background pollution, and (5) mixed industry suggestive of auto repair.

SIGNIFICANCE: Using a community-responsive study design, we characterized ambient VOCs in a pollution-burdened area of Philadelphia. Source apportionment confirmed traffic as a primary VOC source and identified the former refinery site as a source of gasoline and petroleum evaporative emissions, with industrial solvent use as an additional contributor.

IMPACT:

- THRIVEair was a spatially saturated, community-responsive volatile organic compound (VOC) monitoring campaign during the decommissioning and redevelopment of a former refinery site in an industrialized urban area in South Philadelphia. Leveraging data on 38 VOCs, we identified five major VOC sources: (1) vehicular exhaust, (2) gasoline evaporation, (3) industrial solvents, (4) background pollution, and (5) a mixed industrial source suggestive of auto repair. Our study documents air quality impacts of rapid changes in industrial land use at a spatial and temporal scale not previously available in this under-resourced urban community.

Keywords: Air pollution; Environmental justice; Volatile organic compounds, Environmental monitoring

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INTRODUCTION

Urban populations are exposed to a wide range of volatile organic compounds (VOCs) with a range of health impacts from sources including local fossil fuel combustion, industrial sources, and long-range transport from other urban areas. VOCs can impact health as both carcinogens and non-carcinogens, contributing to several adverse health risks [1, 2]. Many VOCs that are common in the urban environment have detrimental health impacts, even at low ambient levels. For example, benzene is a known carcinogen, while its

derivatives toluene, ethylbenzene, and (*m,p*)-xylenes (BTEX) have been linked to respiratory and immune-related diseases, including asthma [3]. In the United States (US), several VOCs are designated as Hazardous Air Pollutants, which allows for regulation of point source emissions by the Environmental Protection Agency (EPA) under the Clean Air Act [4]. However, the EPA has not set health-based ambient exposure limits for VOCs.

In urban communities that border industrial areas (often called “fenceline” communities) [5], air pollution as a whole is associated

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with asthma [6], certain cancers [7], and mental health and quality of life concerns [8, 9]. Like other air pollutants, ambient VOCs tend to be elevated in fenceline communities [10], contributing to elevated health risks [11]. Fenceline community members are also more likely to be economically burdened and people of color, highlighting the environmental justice implications [12]. In Philadelphia, the largest city in Pennsylvania, a 1300-acre swath of land on the Schuylkill River was home to petroleum storage and refining operations for over 150 years [13], while the city grew around it, with a disproportionately Black and low-income fenceline community due to discriminatory housing policies [14].

Despite decades of activism demanding environmental monitoring in communities surrounding the refinery, community air quality remained largely uncharacterized, save for a 2013–2015 monitoring campaign which found elevated concentrations of ambient BTEX surrounding the refinery [15, 16]. In June 2019, a catastrophic explosion caused the refinery, then owned by Philadelphia Energy Solutions (PES), to cease operations and declare bankruptcy. Benzene monitoring on the perimeter of the refinery site, in accordance with the EPA's 2015 Petroleum Refinery Sector Rule [17], shows that the PES refinery was one of the top two benzene emitters of all US refineries in 2019, 2020, and even in 2021, two years after the explosion and during decommissioning [18]. Benzene emissions from PES exceeded the EPA action level of $9 \mu\text{g}/\text{m}^3$ moving annual average from 2018 until perimeter monitoring ended in early 2022 (Fig. S1) [19].

Today, the site, which comprises 2% of Philadelphia's land mass, is being decommissioned and redeveloped into a logistics and warehousing center. Though refinery operations have ceased, local community members have expressed concern about fugitive toxic air emissions during refinery decommissioning and redevelopment. Responding to this concern, we partnered with Philly Thrive to complete a one-year VOC monitoring campaign. Philly Thrive, a local environmental justice organization, has been at the heart of community advocacy around refinery operations, decommissioning, and redevelopment [20]. Through a community-based participatory study design process, we identified and collected samples from 20 air monitoring sites and co-created and disseminated interim findings [21]. In our co-design process, community members identified benzene, lead, and traffic-related air pollutants as primary pollutants of concern, and refinery redevelopment and traffic as major sources of concern. To that end, our community-based report-back process focused on benzene and BTEX. However, we were also able to quantify 38 total VOCs throughout the year, providing an opportunity for additional source identification analyses.

Here, we leverage information on all 38 VOCs to characterize potential sources and their mass contributions in South and Southwest Philadelphia, an industrialized urban area undergoing land use changes. Using validated methods for monitoring a large suite of VOCs using passive methods [22, 23], with a community-responsive spatial saturation approach [21], we employed factor analytic approaches to quantify and compare multiple source contributions during the phase-out of a major refinery in an industrialized urban neighborhood.

MATERIALS AND METHODS

Study design

Partnered with Philly Thrive, we co-designed a VOC monitoring campaign in South and Southwest Philadelphia through an iterative, community-based participatory process; full methodological detail on this co-design process is described elsewhere [21]. Through four monitoring site selection meetings, we identified two primary sources of community concern: a former oil refinery site (in the process of redevelopment into a logistics center) and traffic-related emissions. We selected a total of 20 monitoring sites spanning a range of distances from the refinery, traffic density, and land use (Fig. 1). Due to resource constraints, we were unable to sample all 20 monitoring sites consecutively throughout the year. To ensure that we

could distinguish between major VOC sources in the community while also addressing sites of community concern, we devised a sampling strategy which leveraged a subset of sites to sample year-round and additional sites sampled on a rotating basis, as described below.

Site selection. All potential monitoring sites were required to be sited on a streetside or park utility pole suitable for monitor mounting and owned by the Philadelphia Electric Company (PECO) or the City of Philadelphia and were identified and refined through the same iterative co-design process with Philly Thrive described previously [21].

Year-round sites ($n = 9$): Sites were included in the year-round monitoring network if they met one of the following criteria: (1) representing an expected low-concentration, upwind background location ($n = 1$, Site 1, Bartram's Garden); (2) falling along a transect through, and at increasing distances downwind from, the former refinery site, to isolate the refinery's source contribution ($n = 7$); or (3) co-located with an existing government Air Quality Services (AQS) monitor to allow validation against regulatory monitoring ($n = 1$, Site 8, 24th & Ritner). Several of these locations were themselves proposed by community members in response to specific concerns raised during co-design meetings. For example, the Essington site along the refinery transect was added after community members flagged concerns about the adjacent tank farm.

Rotating sites ($n = 11$): Sites were eligible for inclusion as rotating sites if they were identified by Philly Thrive community members as locations of concern (e.g., schools, parks, or locations with frequently reported odor complaints). From this eligible pool, sites were selected using a stratified sampling approach to ensure representation across [traffic density (low vs. high; 30th/70th percentile of weighted traffic density in 30 m grid cells) [24] and distance to the refinery fenceline (near/mid/far, based on tertiles)], with a minimum of two sites per traffic-by-distance stratum.

To facilitate interpretation aligned with community concern regarding industrial pollution sources, all sites were further classified post hoc as "industrial" or "non-industrial" based on land use within a 100 m buffer, using 2023 tax parcel data from Open Data Philly [25]. Sites were designated industrial if >20% of the buffer area was classified as "Industrial" land use (threshold identified via natural breaks), and non-industrial otherwise.

Sampling schedule. Year-round sites ($n = 9$) were sampled continuously via consecutive one-week samples for the full study duration (July 2023–June 2024). Rotating sites ($n = 11$) were each sampled for two consecutive weeks in summer and two consecutive weeks in winter (four sample-weeks total per site, across two seasonal campaigns). Within each season, sites rotated through in groups of three sampled concurrently for two consecutive weeks before the next group began (e.g., sites 10–12 were sampled together, followed by sites 13–15, with the final group consisting of the remaining two sites).

VOC Sampling and laboratory analysis

Our field team mounted Markes Tenax passive thermal desorption tubes (TDTs) 8–10 feet off the ground on telephone poles in stainless steel weather shelters. TDTs were capped and replaced once per week and refrigerated until they were sent for laboratory analysis (biweekly). We collected one lab blank for each two-week session and one field blank per week and mounted co-located TDTs at one site per week to test reliability.

We used thermal desorption-gas chromatography-mass spectrometry (TD-GC-MS), closely following EPA Method 325B [26], to analyze TDT samples. While EPA Method 325B recommends up to a 14-day sampling period, we selected a one-week sampling duration to achieve finer temporal resolution. Our sampling frequency and experimentally derived seven-day uptake rates have been previously validated [22]. Quality control was ensured using method blanks and a 5-point calibration for all VOC compounds, including routine standard checks using gas standards. Complete laboratory methods are published elsewhere [22, 23]. For each VOC, the limit of detection (LOD) was determined using 3 times the standard deviation of the peak intensity of seven low-level standard measurements, and the upper limit of detection (ULOD) was determined using the largest linear calibration point. GC-MS peaks below the LOD value but above zero were recorded as observations but flagged as LOD; a lack of peak was flagged as non-detect (ND), and the value was replaced with the value of LOD/2; values at or above the ULOD were replaced with the ULOD value. Measurements were subsequently blank-corrected by subtracting the

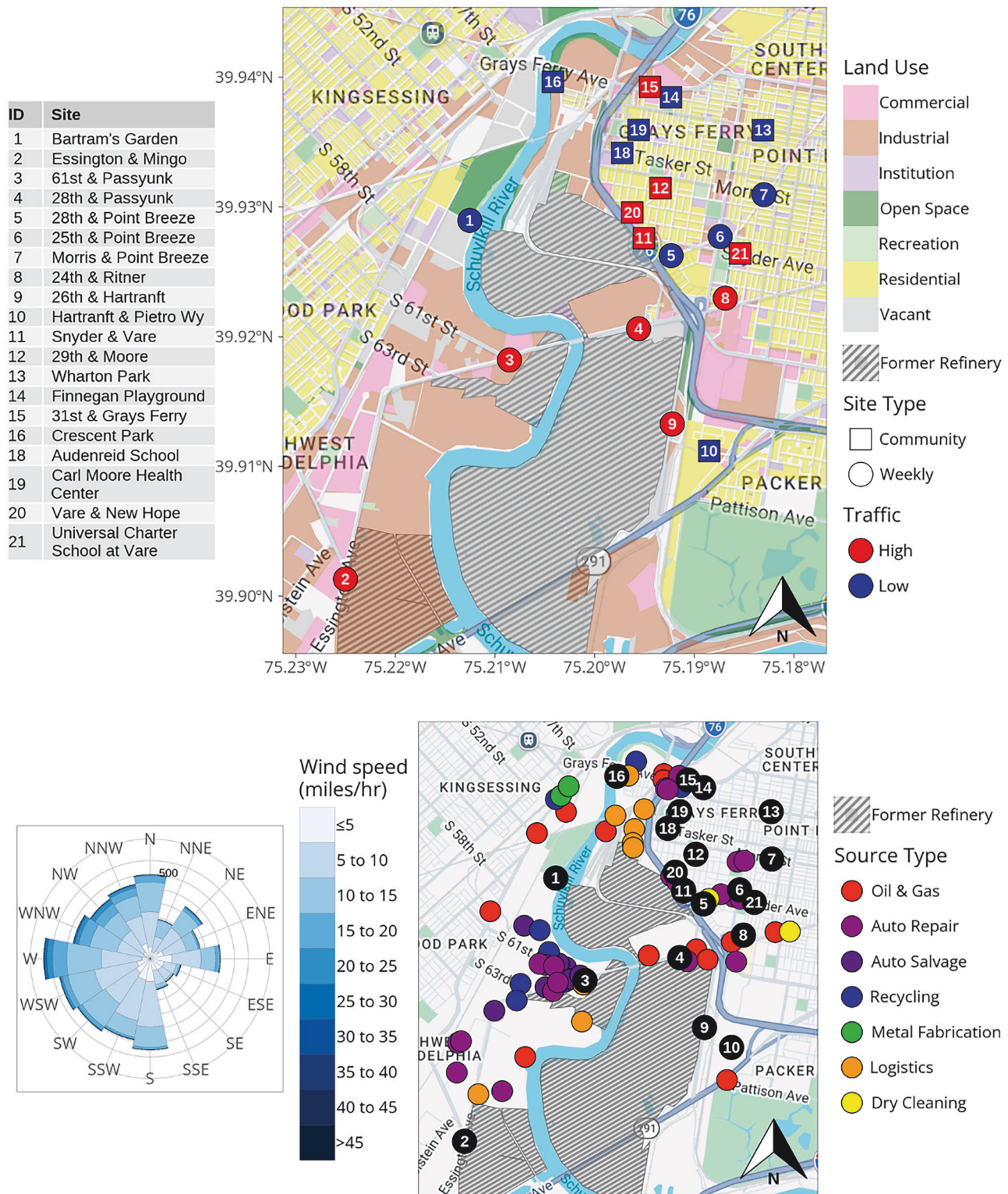


Fig. 1 Monitoring site locations with local land use information (top) and local point sources of pollution (bottom right). Wind rose (bottom left) shows counts (hours) of wind by speed and direction for the Philadelphia Airport weather station from July 1, 2023, to July 1, 2024. Land use parcels were obtained from the City of Philadelphia Department of Planning and Development, and point sources were deduced using Google Maps.

average of the field blanks for that monitoring session. Peak intensities were converted to concentrations using the VOC-specific diffusion rate and the total time the sample was exposed to air [22]. We excluded compounds with 40% or fewer regular (i.e., above LOD and below ULOD) measurements and any compounds with 25% or more non-detects from further analysis.

Statistical analysis

Summary statistics. For each VOC, we calculated the median and interquartile range (IQR) across all sites, within each site, and by season. We visualized VOC correlations using heatmaps of bivariate Spearman rank correlations. Reliability was assessed by calculating the intra-class

correlation coefficient (ICC) and mean relative percent difference (RPD) for duplicate field samples [27]. We excluded compounds with a mean RPD $\geq 50\%$ and/or ICC ≤ 0.75 from further source identification analyses.

Source identification: BTEX ratios. The ratios among benzene, toluene, ethylbenzene, and xylenes (BTEX) concentrations are commonly used to characterize emission sources and/or estimate the photochemical age of BTEX compounds in urban settings [28–33]. Briefly, BTEX ratios can be used to estimate the photochemical age of a BTEX compound, as each species has different reactivity with hydroxyl (OH) radicals (rate constants: benzene < toluene < ethylbenzene < *o*-xylene < *p*-xylene < *m*-xylene) [34]. BTEX ratios can therefore be used to help differentiate sources. Here, we calculated the toluene:benzene (T/B) ratio as a proxy for traffic vs. non-traffic emissions, wherein higher T/B ratios suggest non-traffic emissions [32]. We also report the ratio of (*m* + *p*)-xylenes:ethylbenzene (X/E) as a marker of photochemical age, where higher X/E ratios suggest a local effect, and lower ratios suggest more chemical transport, given greater apparent photochemical degradation [28]. We computed these ratios for each site and by land use and traffic classes.

Source apportionment. To apportion VOC mass across potential sources explaining variation in measured VOC concentrations, we performed Positive Matrix Factorization (PMF) using the EPA PMF 5.0 program [35–37]; details about the PMF program are available in the EPA user manual [38, 39]. We calculated uncertainty for each blank-corrected measurement using relative standard error from field duplicates, uptake rates, and an additional 10% for estimated calibration uncertainty. Measurements flagged as non-detects were assigned an uncertainty of (5/6)*LOD [40].

In the primary PMF analysis, we included all monitoring sites and all monitoring weeks. We excluded compounds from PMF analysis if they met any of the following criteria: (1) < 40% of samples between the lower and upper limit of quantification and > 30% non-detects; (2) poor reliability (ICC ≤ 0.75); (3) Signal-to-noise (S/N) < 0.5. Compounds with S/N greater than 5.0 and less than 1.0 were considered “Weak” and downweighted in the PMF analysis (unless already excluded due to other exclusion criteria). In the PMF software, each observation is assigned a signal-to-noise ratio by subtracting the uncertainty from the observation and dividing by the uncertainty (with negative values set to zero signal). S/N for the species is then the sum of observation-specific S/N divided by the number of observations for that species [39]. We used 20 runs for exploratory analyses and 100 runs for the final analysis. The final number of factors was selected by comparing Q/Q_{exp} with an increasing number of factors, factor stability across bootstrapped samples, and factor interpretability. In PMF, Q is a goodness-of-fit statistic, and Q/Q_{exp} is the ratio of observed vs. expected residual sum of squares. Uncertainty was estimated by pooling 500 bootstrapped samples. Bootstrap median percent contributions were renormalized to sum to 100% per species, since per-factor medians computed independently across bootstrap iterations do not necessarily preserve the species mass balance that holds within each individual run.

Sensitivity analyses. To test the robustness of PMF results to site and week inclusion, we ran PMF restricting to weekly sites only and restricting to summer (June–Aug) and winter (Dec–Feb) to assess the impacts of seasonality.

As an additional sensitivity analysis, we used Principal Component Analysis (PCA) to assess whether VOC source groupings were robust to a different, independent set of modeling assumptions. PCA has been used for VOC source apportionment in several previous studies in urban areas [41–44], but unlike PMF, it does not constrain factor loadings or sample contributions to be non-negative, and it is not subject to the rotational ambiguity inherent to PMF solutions. Comparing the two methods therefore offers a check on whether PMF’s factor structure depends on assumptions specific to that method. A key limitation of PCA is that it forces components to be independent of one another, which is unlikely to be true in an urban environment with mixed sources. We adjusted for within-week correlations by performing PCA on residuals from an intercept-only model with week as a random effect (using a gamma distribution) [45] and weighted the analysis by the total number of observations at each site. Components with an eigenvalue ≥ 1 were retained. We used varimax rotation to improve the interpretability of PCA components while retaining the orthogonality of the components. Component scores for each site were calculated using the residuals from the model at each site and using a weighted sum of component loadings. We ran PCA for all sites and for year-round sites only.

All analyses except for PMF were conducted in R version 4.5.0 [46]. Positive matrix factorization was conducted using the EPA PMF 5.0 software [47]. All R code, data, and results can be accessed at https://github.com/Lfrueh/thriveair_analysis. An interactive data dashboard is available at https://lisa-frueh.shinyapps.io/THRIVEair_results/.

RESULTS

Summary statistics

We were able to detect most compounds with high frequency and with high reliability between paired field samples (Table S1). We excluded the following compounds from further analysis: octane, nonane, 1-hexene, and chloroform (low detection rates); 1,4-diethylbenzene, 1-ethyl-2-methylbenzene, propylbenzene, dodecane, and isoprene (poor/fair reliability); 2-hexanone, methyl isobutyl ketone, and 1,2-dichlorobenzene (both low detection and poor/fair reliability), leaving 26 compounds for further analysis. In general, VOCs were strongly correlated with one another (Fig. S2). Weaker correlations between chlorinated compounds and other VOCs suggest distinct sources of these compounds.

Spatial and temporal variation. Table 1 summarizes the annual median and interquartile range for all VOC species across the entire sampling year. Some seasonal trends were consistent with prior research, with the highest concentrations of BTEX in winter compared to summer (Fig. S3 and Table S3) [48]. A subset of compounds did not show substantial differences in summer vs. winter concentrations (Table S3): all four alkylbenzenes, decane, undecane, 3-methylheptane, 2,3,4-trimethylpentane, and methylcyclohexane. Across sites (Fig. S4), Site 4 (28th & Passyunk) exhibited higher-than-median concentrations of nearly all VOCs, excepting chlorinated compounds and isopentane. Sites 3 (61st & Passyunk) and 9 (26th & Hartranft) exhibited similarly elevated VOC concentrations. All three of these sites are in high-traffic, industrial areas near to the former refinery site. As expected, our background site, Site 1 (Bartram’s Garden) exhibited lower-than-median concentrations of all measured VOCs.

Land use impacts. VOC concentrations differed by land use and traffic categories. Among year-round sites, monitoring sites with $\geq 20\%$ industrial land use in a 100 m buffer around them (“industrial sites”) had higher concentrations of almost all compounds, except for isoprene, carbon tetrachloride, trichlorofluoromethane (CFC-11), and both ketones (Table 1). Among rotating sites, differences between industrial and non-industrial sites may be due to larger measurement variance due to fewer samples. When comparing concentrations by traffic density (Table S2), most VOCs were elevated in high-traffic areas, except isoprene (consistent with biogenic emissions) and nonaromatic chlorinated compounds.

BTEX ratios

BTEX ratios such as the toluene:benzene (T/B) and (*m* + *p*)-xylenes (X/E) ratios can help elucidate VOC sources and photochemical age. In general, we found that BTEX ratios varied by land use and traffic categories (Table 2). We identified higher T/B ratios among industrial sites, compared to non-industrial, when holding traffic constant; among industrial sites, T/B ratios were higher among low-traffic sites. Median T/B ratios varied from 1.62 in high-traffic, non-industrial sites to 1.93 in low-traffic, industrial sites, with an overall median T/B ratio of 1.75, within range of previously-published T/B ratios in urban areas dominated by traffic emissions [36]. Of the year-round sites, Sites 3 (61st & Passyunk) and 6 (25th & Point Breeze) demonstrated higher median T/B ratios than other sites (Table 2). Both these sites are near several auto body shops (Fig. 1), which could explain high toluene concentrations increasing T/B ratios.

Median X/E ratios varied from 6.58 in low-traffic, non-industrial sites to 6.86 in high-traffic, industrial sites, with a study-wide

Table 1. Median [IQR] of each measured VOC for sites in $\mu\text{g}/\text{m}^3$ stratified by site type.

	Year-Round Sites			Rotating Sites		
	Overall 9 sites 408 obs.	Industrial 5 sites 227 obs.	Non-Industrial 4 sites 181 obs.	Overall 11 sites 46 obs.	Industrial 3 sites 12 obs.	Non-Industrial 8 sites 34 obs.
BTEX						
Benzene	1.19 [0.81, 1.59]	1.25 [0.86, 1.65]	1.06 [0.73, 1.53]	1.31 [0.79, 1.59]	1.10 [0.74, 1.48]	1.41 [0.86, 1.61]
Toluene	1.90 [1.45, 2.77]	2.13 [1.53, 3.13]	1.70 [1.33, 2.50]	2.05 [1.40, 2.67]	2.04 [1.73, 2.56]	2.05 [1.40, 2.70]
Ethylbenzene	0.52 [0.39, 0.74]	0.60 [0.41, 0.80]	0.44 [0.36, 0.61]	0.50 [0.39, 0.63]	0.49 [0.44, 0.56]	0.51 [0.37, 0.70]
(m + p)-Xylenes	3.39 [2.52, 4.98]	3.88 [2.94, 5.52]	2.98 [2.32, 4.09]	2.95 [2.21, 4.25]	2.91 [2.64, 3.71]	2.99 [2.06, 4.51]
o-Xylene	0.75 [0.56, 1.07]	0.83 [0.62, 1.21]	0.66 [0.52, 0.87]	0.68 [0.53, 0.91]	0.68 [0.58, 0.78]	0.68 [0.53, 0.96]
Total Xylenes	4.11 [3.12, 6.10]	4.81 [3.58, 6.74]	3.62 [2.90, 4.95]	3.63 [2.73, 5.16]	3.60 [3.31, 4.48]	3.65 [2.61, 5.47]
Alkylbenzenes						
1,4-Diethylbenzene ^a	0.12 [0.07, 0.19]	0.15 [0.08, 0.25]	0.10 [0.07, 0.14]	0.10 [0.07, 0.13]	0.10 [0.07, 0.12]	0.10 [0.07, 0.13]
1,2,3-Trimethylbenzene ^b	0.17 [0.13, 0.26]	0.19 [0.13, 0.31]	0.15 [0.11, 0.20]	0.15 [0.11, 0.18]	0.16 [0.12, 0.18]	0.15 [0.11, 0.18]
1-Ethyl-2-Methylbenzene ^a	0.14 [0.10, 0.20]	0.17 [0.12, 0.26]	0.12 [0.10, 0.16]	0.12 [0.09, 0.15]	0.11 [0.08, 0.14]	0.13 [0.10, 0.16]
1-Ethyl-3-Methylbenzene	0.34 [0.24, 0.51]	0.40 [0.27, 0.64]	0.30 [0.23, 0.41]	0.32 [0.26, 0.40]	0.34 [0.26, 0.35]	0.32 [0.26, 0.40]
1-Ethyl-4-Methylbenzene	0.15 [0.11, 0.24]	0.18 [0.13, 0.31]	0.13 [0.11, 0.18]	0.15 [0.11, 0.19]	0.15 [0.12, 0.17]	0.15 [0.11, 0.20]
Propylbenzene ^a	0.18 [0.12, 0.27]	0.22 [0.15, 0.36]	0.15 [0.12, 0.21]	0.17 [0.15, 0.21]	0.16 [0.12, 0.19]	0.17 [0.15, 0.21]
Mesitylene	0.42 [0.32, 0.68]	0.53 [0.33, 0.87]	0.37 [0.28, 0.49]	0.36 [0.28, 0.44]	0.36 [0.29, 0.41]	0.36 [0.28, 0.48]
C6-C9 Alkanes						
n-Hexane	1.08 [0.66, 1.78]	1.28 [0.78, 2.22]	0.85 [0.59, 1.43]	1.04 [0.55, 1.44]	0.97 [0.53, 1.31]	1.09 [0.58, 1.47]
Heptane	0.83 [0.55, 1.29]	0.97 [0.63, 1.71]	0.66 [0.50, 0.95]	0.77 [0.51, 1.02]	0.74 [0.50, 0.98]	0.77 [0.51, 1.02]
Octane ^a	0.36 [0.23, 0.58]	0.43 [0.26, 0.67]	0.30 [0.20, 0.44]	0.31 [0.20, 0.46]	0.26 [0.17, 0.38]	0.32 [0.21, 0.46]
Nonane ^a	0.53 [0.25, 0.82]	0.56 [0.30, 0.96]	0.46 [0.22, 0.73]	0.46 [0.26, 0.64]	0.43 [0.18, 0.75]	0.50 [0.30, 0.63]
C10-C12 Alkanes						
Decane ^b	0.45 [0.31, 0.74]	0.49 [0.32, 0.92]	0.43 [0.30, 0.56]	0.40 [0.33, 0.51]	0.41 [0.33, 0.47]	0.40 [0.34, 0.51]
Undecane ^b	0.59 [0.38, 0.98]	0.60 [0.41, 1.47]	0.55 [0.37, 0.85]	0.43 [0.37, 0.67]	0.51 [0.42, 0.63]	0.40 [0.35, 0.72]
Dodecane ^a	0.30 [0.18, 0.61]	0.32 [0.20, 0.86]	0.27 [0.16, 0.45]	0.20 [0.14, 0.34]	0.22 [0.18, 0.36]	0.18 [0.13, 0.29]
Other Alkanes						
Isopentane ^b	0.42 [0.28, 0.62]	0.42 [0.33, 0.62]	0.41 [0.24, 0.56]	0.41 [0.14, 0.43]	0.42 [0.14, 0.49]	0.40 [0.14, 0.43]
Isohexane	5.96 [3.70, 10.09]	7.51 [4.28, 13.02]	4.57 [3.20, 7.14]	4.58 [2.74, 7.18]	4.01 [2.61, 4.55]	4.99 [2.93, 8.20]
2-Methylhexane	0.75 [0.50, 1.30]	0.95 [0.58, 1.62]	0.59 [0.41, 0.89]	0.66 [0.41, 0.89]	0.54 [0.50, 0.69]	0.76 [0.39, 0.99]
3-Methylhexane	1.68 [1.14, 2.98]	2.12 [1.34, 3.66]	1.37 [0.94, 1.92]	1.41 [0.95, 1.89]	1.24 [1.15, 1.49]	1.63 [0.91, 2.20]
3-Methylheptane	0.68 [0.44, 1.22]	0.84 [0.53, 1.61]	0.54 [0.38, 0.78]	0.53 [0.42, 0.73]	0.48 [0.41, 0.52]	0.55 [0.45, 0.78]

Table 1. continued

	Year-Round Sites			Rotating Sites		
	Overall 9 sites 408 obs.	Industrial 5 sites 227 obs.	Non-Industrial 4 sites 181 obs.	Overall 11 sites 46 obs.	Industrial 3 sites 12 obs.	Non-Industrial 8 sites 34 obs.
3-Methylpentane	2.09 [1.25, 3.86]	2.80 [1.44, 5.09]	1.57 [1.11, 2.44]	1.58 [1.02, 2.44]	1.38 [0.97, 1.66]	1.73 [1.07, 2.75]
2,4-Dimethylpentane	0.34 [0.23, 0.58]	0.44 [0.27, 0.79]	0.28 [0.20, 0.39]	0.23 [0.19, 0.36]	0.22 [0.18, 0.26]	0.24 [0.19, 0.41]
2,3,4-Trimethylpentane	0.42 [0.28, 0.75]	0.53 [0.33, 1.06]	0.36 [0.25, 0.53]	0.30 [0.23, 0.44]	0.25 [0.23, 0.28]	0.36 [0.25, 0.48]
Methylcyclohexane	1.59 [1.01, 3.12]	2.06 [1.22, 4.51]	1.30 [0.84, 2.00]	1.11 [0.83, 1.57]	0.92 [0.81, 1.10]	1.29 [0.86, 1.69]
Alkenes						
1-Hexene ^a	0.20 [0.11, 0.31]	0.23 [0.14, 0.37]	0.16 [0.09, 0.23]	0.18 [0.10, 0.26]	0.17 [0.14, 0.23]	0.19 [0.10, 0.26]
cis-2-Pentene ^b	0.04 [0.02, 0.08]	0.04 [0.02, 0.09]	0.04 [0.01, 0.06]	0.04 [0.02, 0.07]	0.04 [0.03, 0.07]	0.04 [0.02, 0.07]
Isoprene ^a	0.03 [0.01, 0.10]	0.01 [0.01, 0.09]	0.03 [0.01, 0.11]	0.03 [0.01, 0.06]	0.01 [0.01, 0.04]	0.03 [0.01, 0.06]
Ketones						
2-Hexanone ^a	0.01 [0.01, 0.01]	0.01 [0.01, 0.01]	0.01 [0.01, 0.01]	0.01 [0.01, 0.01]	0.01 [0.01, 0.01]	0.01 [0.01, 0.01]
Methyl isobutyl ketone ^a	0.00 [0.00, 0.04]	0.00 [0.00, 0.04]	0.00 [0.00, 0.04]	0.00 [0.00, 0.04]	0.00 [0.00, 0.01]	0.00 [0.00, 0.04]
Chlorinated Compounds						
Chloroform ^a	0.17 [0.13, 0.20]	0.16 [0.13, 0.18]	0.17 [0.14, 0.22]	0.18 [0.14, 0.21]	0.18 [0.14, 0.21]	0.18 [0.15, 0.22]
Carbon tetrachloride	1.73 [1.26, 2.09]	1.72 [1.26, 2.11]	1.74 [1.26, 2.07]	1.90 [1.37, 2.28]	2.07 [1.25, 2.32]	1.85 [1.68, 2.24]
Tetra-chloroethylene	0.12 [0.09, 0.16]	0.12 [0.09, 0.18]	0.12 [0.09, 0.15]	0.14 [0.09, 0.17]	0.18 [0.13, 0.19]	0.12 [0.09, 0.15]
Trichloro-fluoromethane	0.61 [0.31, 0.92]	0.61 [0.32, 0.92]	0.59 [0.29, 0.91]	0.68 [0.34, 1.21]	0.62 [0.32, 1.19]	0.68 [0.40, 1.21]
1,2-dichlorobenzene ^a	0.11 [0.07, 0.16]	0.09 [0.05, 0.12]	0.15 [0.11, 0.22]	0.14 [0.09, 0.19]	0.09 [0.09, 0.15]	0.15 [0.12, 0.20]

^aExcluded from principal component analysis and positive matrix factorization due to poor reliability or low detection rates (see Table S1).

^bDown weighted in positive matrix factorization analysis due to poor signal:noise ratio.

Year-round sites were sampled weekly for one year, and rotating sites were sampled for 4 total weeks in a rotating fashion across one summer and one winter session. Sites were considered industrial if >20% of the area within a 100 m buffer of the site is of industrial land use categorization.

median of 6.72. X/E ratios were higher in industrial sites compared to non-industrial sites (Table 2). Of the year-round sites, Site 4, which had the highest concentration of most VOCs compared to other sites, consistently exhibited the highest median X/E ratio (7.56 compared to the study-wide median of 6.72) compared to other sites (Table 2). This site had the highest X/E ratios across the year, except for one three-week period (Fig. S5).

In the three sampling weeks spanning October 25th through November 8th, 2023, Site 4 showed extremely low T/B (close to zero) and X/E (close to 2) ratios (Fig. S5), due to high benzene and ethylbenzene concentrations, respectively, during these weeks. Other nearby sites did not reflect the same pattern, suggesting a highly localized source of benzene and ethylbenzene during these weeks. Among the sites sampled weekly, our background site, Site 1 (Bartram's Garden), had the lowest T/B and X/E ratios, consistent with fewer fresh emission sources nearby.

Source apportionment

In addition to excluding the same compounds for poor reliability and/or low detection rates, we down-weighted the following compounds with $0.5 \leq S/N < 1.0$ in PMF analysis: 2-pentene, 1,2,3-trimethylbenzene, isopentane, undecane, and decane (Table 1). By

comparing Q/Q_{exp} across 2 to 6 factors, we found no clear inflection point indicating the ideal number of factors, therefore we selected a 5-factor solution with $F_{\text{peak rotation}} = 0.5$ based on stability of factors across bootstrapped samples and factor interpretability. Figure 2 shows the normalized contribution of each VOC to each PMF factor, and Fig. 3 shows the contribution of each factor to the total mass of each VOC. Isohexane is ubiquitous in all five factors, suggesting multiple sources of this compound. Figure 4 shows the contribution of each factor to total VOC mass at each site. From these results, we interpreted the five-factor solution as follows.

Vehicle exhaust. The factor explaining most of the total VOC mass (36%) consists of BTEX, hexane, and heptane (Fig. 2). BTEX, particularly toluene, is indicative of gasoline exhaust [49, 50], and cis-2-pentene is an indicator of incomplete combustion [49, 51]. The vehicle exhaust factor contributed ~10-25% of the mass for most compounds (Fig. 3), except for the chlorinated compounds (which we link to background pollution and industrial solvents), isopentane, and 2,3,4-trimethylpentane, which are both enriched in liquid gasoline and gasoline evaporation [49]. We saw some variation in the contribution of the vehicle exhaust factor to total

Table 2. Annual median [IQR] of benzene, toluene, ethylbenzene, and xylenes (BTEX) concentrations in $\mu\text{g}/\text{m}^3$ and BTEX ratios by traffic and industrial site category, and across year-round sites.

		T/B	X/E	B	T	E	(m,p)-X	o-X
All Sites								
All Sites		1.75 [1.47, 2.12]	6.72 [5.99, 7.28]	1.21 [0.81, 1.59]	1.91 [1.44, 2.76]	0.52 [0.39, 0.73]	3.37 [2.51, 4.92]	0.74 [0.56, 1.03]
Industrial	HighT	1.80 [1.41, 2.23]	6.86 [6.07, 7.48]	1.27 [0.90, 1.76]	2.13 [1.54, 3.19]	0.61 [0.45, 0.83]	4.13 [3.03, 5.91]	0.89 [0.65, 1.30]
	LowT	1.93 [1.71, 2.54]	6.75 [6.21, 7.30]	1.05 [0.67, 1.45]	2.10 [1.50, 2.92]	0.47 [0.38, 0.65]	3.38 [2.53, 4.48]	0.69 [0.56, 0.88]
Non-Industrial	HighT	1.62 [1.32, 1.93]	6.58 [5.94, 7.23]	1.34 [0.86, 1.72]	1.90 [1.51, 2.53]	0.52 [0.42, 0.62]	3.32 [2.54, 4.25]	0.71 [0.60, 0.91]
	LowT	1.70 [1.48, 1.98]	6.59 [5.96, 7.08]	1.05 [0.71, 1.51]	1.69 [1.29, 2.54]	0.43 [0.33, 0.61]	2.78 [2.19, 4.08]	0.63 [0.48, 0.87]
Year-Round Sites								
Year-Round Sites								
Site 1		1.75 [1.48, 2.14]	6.80 [6.08, 7.34]	1.19 [0.81, 1.59]	1.90 [1.45, 2.77]	0.52 [0.39, 0.74]	3.39 [2.52, 4.98]	0.75 [0.56, 1.07]
Site 2		1.69 [1.45, 1.93]	6.40 [5.78, 6.70]	0.91 [0.64, 1.19]	1.36 [1.00, 1.95]	0.33 [0.26, 0.46]	2.12 [1.46, 2.81]	0.48 [0.36, 0.63]
Site 3		1.91 [1.70, 2.22]	6.60 [5.97, 7.31]	1.07 [0.73, 1.38]	1.87 [1.48, 2.66]	0.45 [0.35, 0.58]	3.16 [2.28, 3.74]	0.66 [0.50, 0.80]
Site 4		2.36 [2.15, 2.89]	6.89 [6.40, 7.29]	1.36 [0.93, 1.85]	2.96 [2.13, 4.76]	0.77 [0.61, 1.27]	5.38 [3.88, 8.66]	1.17 [0.99, 1.81]
Site 5		1.49 [1.28, 1.82]	7.56 [6.57, 8.43]	1.50 [1.10, 2.16]	2.36 [1.45, 3.60]	0.77 [0.53, 1.67]	5.40 [3.95, 12.37]	1.19 [0.83, 2.74]
Site 6		1.74 [1.49, 2.03]	6.87 [6.12, 7.35]	1.12 [0.72, 1.57]	1.82 [1.38, 2.80]	0.48 [0.39, 0.72]	3.21 [2.52, 4.84]	0.69 [0.58, 0.98]
Site 7		2.02 [1.71, 2.54]	6.88 [6.40, 7.40]	1.05 [0.67, 1.50]	2.10 [1.55, 3.07]	0.47 [0.38, 0.69]	3.42 [2.55, 4.63]	0.71 [0.57, 0.97]
Site 8		1.65 [1.49, 1.94]	6.76 [6.33, 7.11]	1.23 [0.82, 1.53]	1.74 [1.46, 2.59]	0.48 [0.40, 0.63]	3.14 [2.48, 4.40]	0.70 [0.55, 0.90]
Site 9		1.65 [1.32, 1.97]	6.69 [5.98, 7.31]	1.29 [0.86, 1.72]	1.89 [1.53, 2.59]	0.52 [0.43, 0.67]	3.40 [2.68, 4.88]	0.72 [0.63, 1.02]
Site 9		1.49 [1.27, 1.65]	6.75 [6.00, 7.14]	1.20 [1.04, 1.56]	1.79 [1.46, 2.29]	0.56 [0.46, 0.68]	3.64 [2.84, 4.43]	0.79 [0.63, 0.98]

HighT High Traffic, LowT Low Traffic, T/B toluene:benzene ratio, X/E (m,p)-xylenes:ethylbenzene ratio.

VOC mass across sites (Fig. 4), with total contributions ranging from 13% at site 9 to 25% at site 18. However, when comparing sites classified as high- vs. low-traffic, there was no clear pattern in total VOC mass contribution by this source. These findings suggest that either our traffic categorization, based on weighted traffic density, did not appropriately capture traffic emissions, or that vehicle exhaust is a ubiquitous exposure in our study area.

Gasoline evaporation. The factor accounting for 27% of the total VOC mass was characterized by methylcyclohexane, branched alkanes, and 2,3,4-trimethylpentane, an isomer of octane. All of these compounds are enriched in gasoline, but given the absence of combustion indicators, we interpret this factor as gasoline/petroleum evaporation [49].

The gasoline evaporative emissions source exhibited the most dramatic spatial variation across sites (Fig. 4), ranging from 0% at sites 12, 14, and 15 to 44% at site 4. This suggests highly localized emissions, perhaps capturing petroleum evaporation released during construction activities on the former refinery site. This source contributed close to zero VOC mass to sites in residential areas (e.g., site 12) and in parks (e.g., sites 1, 13, 14, 16). Conversely, the evaporative emissions source accounted for 25–50% of the total VOC mass at sites, 44% of the total VOC mass at site 4 and 39% at site 9, which border the former refinery site. Even among sites bordering industrial areas, large spatial variation was observed—the source contributed very little VOC mass at site 2, which borders the former refinery site on its west side and sits along a major road. Notably, this site sits upwind of the former refinery site (Fig. 1). When comparing only sites sampled weekly (sites 1–9), sites 5, 6, and 7 form a downwind transect from the former refinery site—the petroleum evaporative emissions source contributed 16%, 9%, and 7% of VOC mass at these sites, decreasing as distance from the refinery fence line increases.

Industrial solvents. Contributing 14% of the total mass, the industrial solvent factor is characterized by 1,2,3-trimethylbenzene, decane, and undecane. Additional key compounds include carbon tetrachloride, (m,p)-xylenes, and alkylbenzenes, while shorter branched alkanes are missing from this factor. Aromatic hydrocarbons such as 1,2,3-trimethylbenzene and BTEX are found in industrial solvents [52] and in motor vehicle exhaust [49]—because of the absence of combustion indicators and short-chain branched alkanes that were abundant in the vehicle exhaust factor, we conclude that this factor is due to industrial hydrocarbon solvent use.

Background pollution. Explaining 12% of the total VOC mass, we interpreted the factor characterized by carbon tetrachloride and trichlorofluoromethane (CFC-11) to be background pollution. Carbon tetrachloride and CFC-11 are both ozone-depleting substances that were phased out after the United Nations' 1989 Montreal Protocol [53]. Nevertheless, contemporary sources of carbon tetrachloride persist, including chlorine production and usage, and CFC-11 persists in the atmosphere, in part due to improper disposal of appliances containing the compound [54, 55]. Notably, there was little spatial variation in the background pollution source, with similar temporal patterns across sites, suggesting long-range transport rather than a localized source.

Mixed industry. We identified a mixed industry factor characterized by isohexane and tetrachloroethylene, explaining 11% of the total VOC mass. Isohexane, toluene, and (m,p)-xylenes were also loaded onto this factor. Isopentane is enriched in evaporated gasoline (compared to vehicle exhaust) [49], and at the time of this study tetrachloroethylene was widely used as a component in metal degreasers and brake cleaners [56]. Given the high density

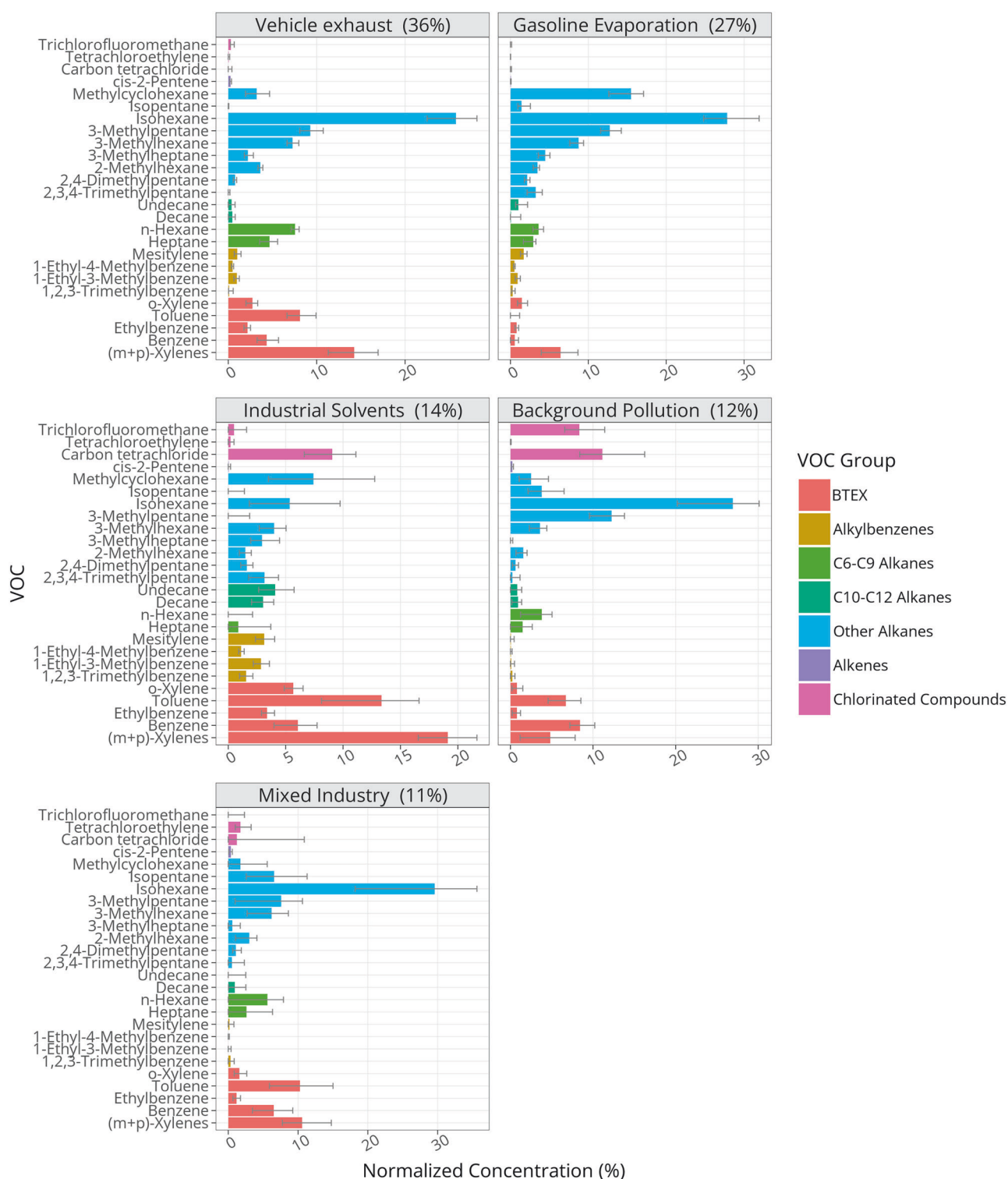


Fig. 2 Concentration contribution (%) of each compound to PMF-derived factors, normalized to the median of 500 bootstrapped samples. Colors correspond to broad chemical classes of volatile organic compounds (VOCs). Error bars showing the 5th and 95th percentiles of the bootstrapped distribution. The proportion of total predicted mass explained by each factor is shown in parentheses.

of auto repair shops (Fig. 1) in the study area, it is possible that this source indicates auto repair emissions.

Consistent with the spatial co-occurrence of sources in South Philadelphia, sources related to industry (gasoline evaporation, solvent use, and mixed industry) contributed to roughly 50% of total VOC mass at all 20 sites, regardless of surrounding land-use designation (Fig. 4), though industrial sites had a slightly higher

average industrial contribution (61% vs. 55% for non-industrial sites). Vehicular exhaust was the other largest contributor to VOC mass (36%).

Sensitivity analyses. PMF analyses restricted to summer (June–August) and winter (December–February) yielded similar results to analyses including the entire calendar year (Fig. S11),

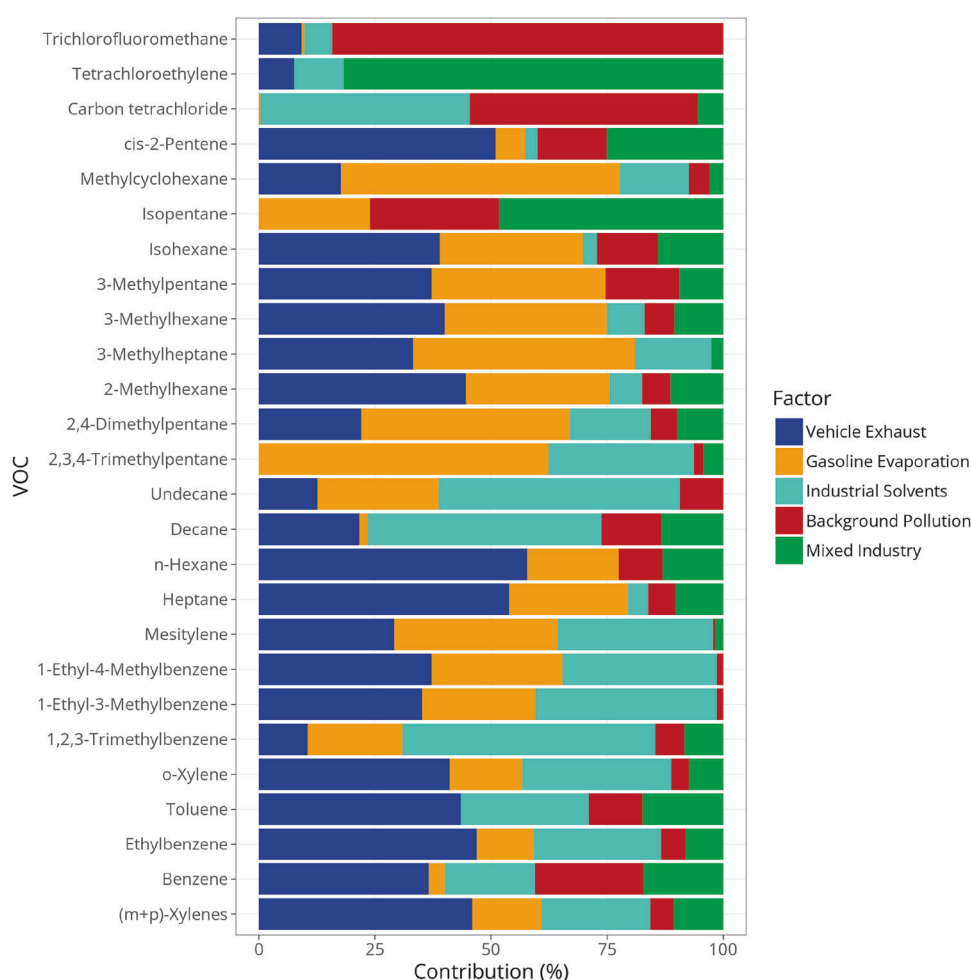


Fig. 3 Percent of each VOC species mass explained by each factor in positive matrix factorization (PMF) analysis. Colors correspond to factors in the 5-factor solution.

though a four-factor solution was selected over a five-factor solution due to stability across bootstrapped samples. In both season-restricted analyses, the “background pollution” and “mixed industry” factors from the primary analysis were collapsed into one combined factor, as evidenced by CFC-11 co-occurring with toluene, xylenes, and hexane (Figs. S9 and S10). A plausible explanation for this change is statistical power: With roughly one-quarter of the sample size available in each season-restricted analysis compared to year-round, PMF may have insufficient data to resolve the two smallest-mass factors that were least robustly separated to begin with. A key difference between the two season-restricted analyses is that 2,3,4-trimethylpentane mass was primarily explained by the gasoline evaporation factor in winter and in the primary analysis, while in summer, the mass was primarily explained by the vehicular exhaust/combustion factor. This shift may reflect the temperature-dependent nature of evaporative emissions. Because 2,3,4-trimethylpentane is enriched in gasoline vapor due to its high vapor pressure, its evaporative emission rate is expected to be greater in warmer conditions. Outside of this discrepancy, summer and winter VOC contributions to factor masses were very similar to the primary analysis for the vehicular exhaust, gasoline evaporation, and industrial solvent factors, indicating that our source apportionment results are robust to seasonal effects. When PMF analysis was repeated for year-round sites only, results were nearly identical to the main analysis (Fig. S7).

Principal Component Analysis (PCA) confirmed that traffic is a significant contributor to VOCs in our study area—we interpreted

Component 1, explaining 64% of the variance, as a vehicular exhaust source owing to the loading of *cis*-2-pentene, alkylbenzenes, and BTEX (Fig. S12). PCA Component 2, explaining 12% of variance, identified compounds relating to gasoline evaporation (e.g., methylcyclohexane and branched alkanes), consistent with PMF results, though 2,3,4-trimethylpentane loaded weakly negatively to this component, inconsistent with PMF results. Like PMF, PCA identified a background pollution source with CFC-11 and carbon tetrachloride (Component 3 in Fig. S12). In the PCA results, the mixed industry and industrial solvent sources were grouped in the same component, albeit with opposite signs (Component 4 in Fig. S12). We interpret this divergence as evidence that these two sources, while both contributing meaningfully to total VOC mass, are spatially or temporally anti-correlated, consistent with geographically distinct point sources (e.g., separate auto repair or solvent-use locations) rather than a single diffuse industrial source artificially split into two factors by PMF. PCA results were unchanged when restricted to year-round sites only (Fig. S13).

DISCUSSION

The THRIVEair study was a community-based air monitoring project in S/SW Philadelphia from summer 2023 to summer 2024. Following community-based participatory research methods, we co-designed our air monitoring network and report-back materials and presentations with Philly Thrive, a local environmental justice organization [21]. We measured VOCs for one year, using highly

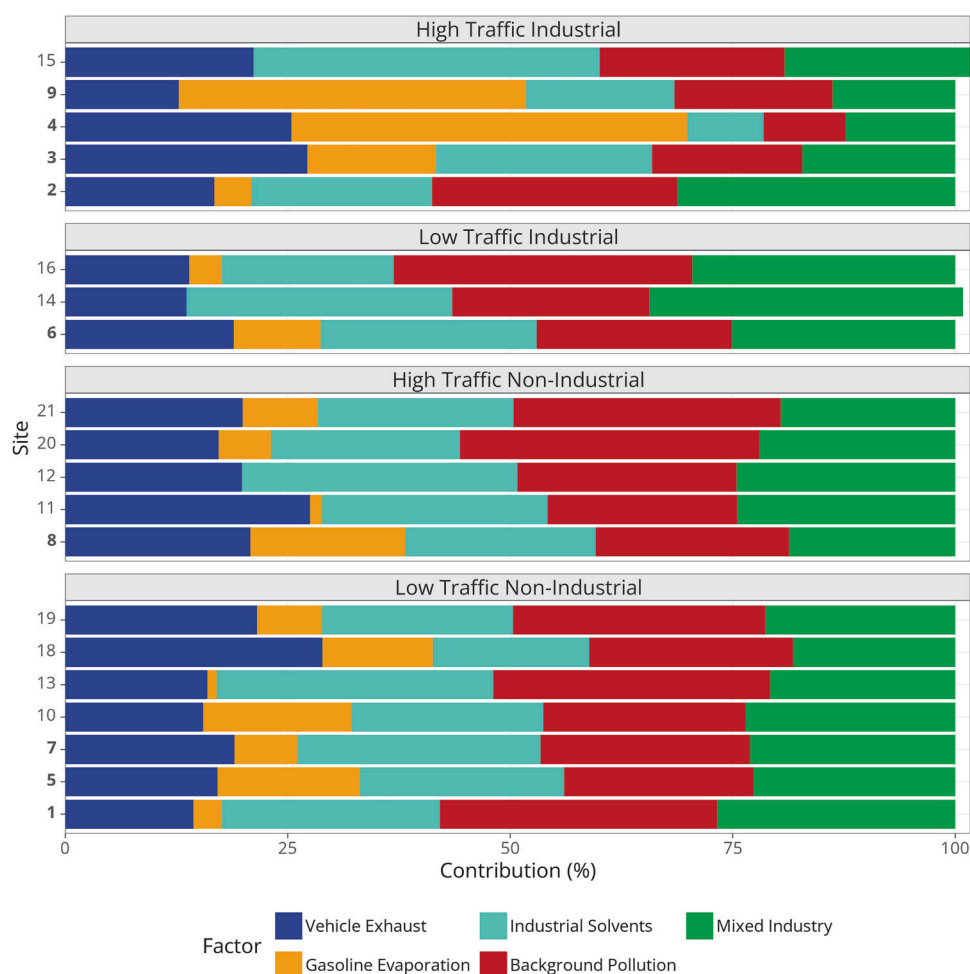


Fig. 4 Positive matrix factorization (PMF) factors' contribution to total VOC mass at each site, by land use categories. Year-round site IDs are shown in **bold**. Colors correspond to factors in the 5-factor PMF solution.

sensitive analytic methods allowing for the detection of 38 VOCs at low concentrations. While we focused only on benzene, toluene, ethylbenzene, and xylenes (BTEX) in our community-facing report-backs, in this analysis, we sought to examine groupings of VOCs using multivariate methods to identify local VOC sources and source contributions.

BTEX concentrations in context

Through our co-design process, community members reported being most concerned about benzene, toluene, ethylbenzene, and xylenes (BTEX) emissions from traffic and from decommissioning and redevelopment at a former refinery site. Throughout the year, we measured low benzene concentrations (mean = $1.34 \mu\text{g}/\text{m}^3$, median = $1.21 \mu\text{g}/\text{m}^3$, range = 0.31 to $9.05 \mu\text{g}/\text{m}^3$) consistent with previously published concentrations in urban areas in Europe and the Americas, with mean benzene concentrations below $2 \mu\text{g}/\text{m}^3$ (Table S4) [22, 57–60].

Compared to a previous passive VOC sampling study conducted in the same area during active refinery operations from 2013–2015 [15], we measured lower benzene concentrations during our 2023–2024 campaign. While the previous study used slightly different methods (2-week integrated samples vs. 1-week integrated samples), we believe our design to be comparable. Though we measured lower average concentrations of benzene in our campaign, concentrations of toluene, ethylbenzene, and (*m,p*)-xylenes were similar to those during active refinery operation, suggesting other prominent sources in the area. We also observed some concerning peaks—the maximum (*m,p*)-xylenes

concentration was $37 \mu\text{g}/\text{m}^3$. Notably, we measured average (*m,p*)-xylenes concentrations that were higher than those reported in most other urban studies (Table S4)—our study-wide average was $4.6 \mu\text{g}/\text{m}^3$, much higher than other North American urban studies [22, 57, 58], and aligning more with mixed-use and industrial sites in large cities such as La Plata, Argentina [61] and Ulsan, South Korea [62]. Much of these high concentrations were driven by extreme values at a single site (Site 4), though the study-wide average excluding this site remains high ($4.0 \mu\text{g}/\text{m}^3$) when compared to similar urban environments. We also reported high (*m,p*)-xylenes:ethylbenzene (X/E) ratios (> 6.0) when compared to other studies (range 0.9–4.3 among studies included in Table S4). The observed elevated (*m,p*)-xylenes concentrations in our study could be due to the high concentration of auto body shops in the study area (Fig. 1) since xylene isomers are commonly emitted from auto paints, primers, degreasers, and solvents [63].

BTEX ratios as source signatures

We found that BTEX ratios, which have been used by several studies to understand source signatures and photochemical aging, were less informative for source identification in this study.

Previous literature has identified a T/B ratio of 2.0 to indicate traffic emissions, with ratios > 2.0 suggesting volatilization (i.e., industrial sources) [36, 64]. Other studies have used a threshold of 3 [62], 2.2 [65], or a range of 1.5 to 4.0 [32] to suggest traffic emissions. Most studies cite that very low T/B ratios indicate biomass burning [36, 65]. In our review of the previous literature, we found no clear threshold of T/B ratio delineating industrial vs.

background urban emissions (Table S4). We measured toluene:benzene (T/B) ratios (study-wide mean = 1.82) that were consistent with studies in other industrialized cities, including in the same community when the refinery was operational in 2016 [57]. Generally, we found higher toluene:benzene (T/B) ratios in industrial areas compared to non-industrial areas, consistent with fresh petroleum sources (Table 2). Similar patterns were reported in Kim et al.'s 2019 study in Ulsan [62] and Hoque's 2008 study in Delhi [66]. Conversely, Lerner et al. reported the lowest T/B ratios in industrial sites compared to semiurban and urban sites in La Plata [61], and there were no differences in T/B ratios when comparing sites near and far from the active PES refinery in Philadelphia in 2015 [15].

A second widely used BTEX ratio is the (*m,p*)-xylenes:ethylbenzene (X/E) ratio, which indicates the photochemical age of the air mass, with higher ratios indicating fresh sources and lower ratios indicating chemical aging [28, 36]. We found that (*m,p*)-xylenes:ethylbenzene (X/E) ratios varied across individual sites (study-wide mean = 6.7, site annual mean range = 2.2 to 10.5), yet there were few differences by land use categories (Table 2). Among sites sampled weekly, we found the lowest X/E ratios at Site 1 (Bartram's Garden), a background site away from fresh source emissions, and the highest ratios at Site 4, a high-traffic industrial site near an oil terminal (Table 2). In October and November of 2023, a three-week dramatic dip in both T/B and X/E ratios Site 4 (Fig. S5), without a similar pattern among nearby sites. During this window, the T/B ratio dipped to nearly zero, followed by two additional weeks below 0.8 at the site, compared to normal ratio values between 1 and 3 at that site. The X/E ratio dipped below 3.0 at the same time, though X/E ratios remained between 5 and 10 throughout the rest of the year. These dips corresponded to peaks in benzene and ethylbenzene concentrations at the site, suggesting a highly localized emission event, likely on the former refinery site.

Spatial VOC patterns

Out of the 20 sites we monitored, three sites stood out with high VOC concentrations across multiple VOC categories (Fig. S3): Sites 4, 3, and 9 (28th & Passyunk, 61st & Passyunk, and 26th & Hartranft). All three sites border the former refinery site (Fig. 1), sit on major roads with significant diesel truck traffic, and lie adjacent to active industrial areas. In comparison, our rotating sites, located mostly in residential areas, showed lower concentrations of most VOCs. Site 4 showed consistently higher (in some cases, dramatically higher) VOC concentrations compared to all other sites, even those nearby, highlighting localized emissions. For example, the mean (*m,p*)-xylenes concentration at Site 4 was 10.4 $\mu\text{g}/\text{m}^3$, far exceeding the study-wide average of 4.6 $\mu\text{g}/\text{m}^3$, and 65% higher than the next-highest average site. The maximum concentrations for all BTEX species at Site 4 were 1.5 to 4.5 times higher than the maxima at any other site. For the four BTEX species, 1 benzene, 4 toluene, 2 (*m,p*)-xylenes, and 1 *o*-xylene measurement exceeded the upper limit of detection (ULOD)—all of these, save for two ULOD toluene measurements, occurred at Site 4. While the former refinery site is no longer active, our findings at Site 4, along with the temporal patterns of BTEX ratios, are suggestive that refinery decommissioning and construction activities had highly localized impacts on air quality.

Seasonal VOC patterns

Consistent with prior research in temperate urban environments, we observed higher BTEX concentrations in winter compared to summer (Table S3) [48]. This pattern is commonly attributed to a combination of factors that suppress dispersion and chemical removal in winter, including lower atmospheric boundary layer heights, reduced photochemical activity, and added wintertime combustion from residential heating. A subset of compounds—all four included alkylbenzenes, decane, undecane, 3-methylheptane,

2,3,4-trimethylpentane, and methylcyclohexane—did not show substantial seasonal differences (Table S3). This contrast becomes more interpretable when considered alongside our source apportionment results. Decane, undecane, and the alkylbenzenes were primary contributors to the industrial solvents factor, while methylcyclohexane and 2,3,4-trimethylpentane were key compounds related to the gasoline evaporation factor. Both of these factors exhibited dramatic spatial variability (Fig. 4), consistent with localized point sources (e.g., ongoing construction activity at the former refinery site), suggesting greater spatial, compared to temporal, variability.

Source apportionment profile

Source apportionment using PMF suggested five VOC sources, from most to least abundant: (1) vehicular exhaust, (2) gasoline evaporation, (3) industrial solvents, (4) background pollution, and (5) mixed industrial source suggestive of auto repair. Most VOCs loaded onto more than one component or factor (Fig. 2), highlighting some ambiguity in factor interpretation that is common among urban VOC source apportionment studies with spatially co-located urban and industrial pollution sources [67].

The *vehicle exhaust* source was identified by BTEX [49], hexane, and heptane, indicators of vehicle exhaust, and *cis*-2-pentene, an indicator of incomplete combustion [49, 51]. These indicator compounds are commonly identified with vehicle exhaust sources [49, 68, 69], and similar VOC fingerprints for this source have been identified in source apportionment studies, for example, in Los Angeles [70] and Alberta, Canada [71]. The *gasoline evaporation* source was defined by the presence of methylcyclohexane, 2,3,4-trimethylpentane [49, 72], light branched alkanes, and the absence of combustion indicators. Together, these indicate an evaporative emissions source [69, 71]. The *industrial solvents* source was defined by 1,2,3-trimethylbenzene, decane, undecane, and carbon tetrachloride, with the absence of shorter branched alkanes. In Los Angeles, Brown et al. similarly identified C9-C11 alkanes along with toluene as indicating industrial coatings and solvents [70]. The *background pollution* source was defined by CFC-11 and carbon tetrachloride, which are both very stable in the lower atmosphere and have been phased out of active use [54, 55]. Other source apportionment studies in North America have identified these compounds as indicators of background pollution [71, 73]. The *mixed industry* source was defined by tetrachloroethylene [56], a recently banned compound commonly used in automotive solvents and degreasers, and isopentane, an indicator of gasoline evaporation [49]. Because various point sources of VOC emissions are highly spatially clustered in the study area (Fig. 1), we were unable to further specify individual industries as VOC sources. Nevertheless, our overall source profile aligns with other source apportionment studies in industrialized areas, which distinguish between gasoline vehicles (combustion), gasoline evaporation, and industrial emissions [49], for example, in Pittsburgh, Pennsylvania [74], Baltimore, Maryland [75], and Los Angeles, California [70].

PMF restricted to year-round sites alone produced similar results to the primary analysis. Season-restricted PMF analyses (summer and winter) also broadly reproduced the primary source profiles, though a four-rather than five-factor solution was preferred in both cases: the background pollution and mixed industry factors collapsed into one combined factor, likely reflecting reduced statistical power. As an additional sensitivity analysis, we used principal component analysis (PCA) to assess whether this factor structure was robust to different modeling assumptions. PCA broadly supported the PMF solution, but retained only four components rather than five: the mixed industry and industrial solvents factors loaded onto a single PCA component with opposing signs, rather than mapping to separate components as the other PMF factors did. This suggests an anti-correlation between the mixed industrial and industrial solvent

sources, suggesting spatial and/or temporal separation. Overall, the consistency of the vehicular exhaust, gasoline evaporation, and industrial solvents factors across site subsets, seasons, and models suggests that our primary source apportionment results reflect spatially and temporally stable sources, rather than artifacts of a particular sampling period or site selection.

Strengths and limitations

This study has several notable strengths. We characterized a large number of VOCs (38 detected and 26 retained in source apportionment analyses) in a spatially saturated, community-responsive design. Our one-week integrated sampling approach allowed for detection of VOCs at very low concentrations, and our community-based participatory co-design process ensured that the results had direct relevance for community partners.

Our study also has several limitations. First, due to our relatively small sample size and unbalanced sampling design, we were unable to perform land use regression or other spatial interpolation methods with confidence. Second, we could not reliably collect short-chain (C2–C5) hydrocarbons using the Tenax sorbent. These short-chain hydrocarbons could help refine our source apportionment, particularly in distinguishing combustion- from evaporation-related sources. Third, because we only sampled across one calendar year, it is unclear whether temporal variations in VOC concentrations are truly seasonal or unique to the sampling year. Finally, PMF solutions are subject to rotational ambiguity. While we assessed factor stability using bootstrapping and Fpeak rotation and confirmed our overall factor structure was robust to an independent multivariate method (PCA) and to restricting our sample to year-round sites or individual seasons, some ambiguity in factor interpretation remains. However, this is an inherent limitation of PMF source apportionment in an area with spatially co-located emission sources.

CONCLUSION

Historically disinvested neighborhoods in South and Southwest Philadelphia experience a mix of air pollution sources, from traffic, industrial areas, the petroleum industry, and recent construction on a former refinery site. Through a community-responsive air monitoring network, we characterized 38 VOCs across 52 weeks and 20 monitoring sites. Source apportionment analyses corroborated sources of community concern, namely traffic, while highlighting industrial solvent use as an additional contributor to local VOC pollution. The combination of high-resolution temporal sampling, broad VOC coverage, and a community-driven site selection process allowed us to characterize ambient VOC concentrations and sources in a pollution-burdened community with a level of spatial and chemical detail not previously available for this area.

DATA AVAILABILITY

All data are available at https://github.com/Lfrueh/thriveair_analysis.

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AUTHOR CONTRIBUTIONS

Conceptualization: LF, KM, JEC, NACJ and ST; Data curation: LF, KM, GT, KCW, LJ, NACJ and ST; Formal analysis: LF; Funding acquisition: LF, JEC, NACJ and ST; Investigation: LF, KM, GT, KCW, LJ, NACJ and ST; Methodology: LF, KM, NACJ and ST; Project administration: JEC, NACJ and ST; Resources: ST; Software: LF, LJ and KM; Supervision: JEC, NACJ and ST; Validation: LF; Visualization: LF; Writing—Original Draft: LF and ST; Writing—Review and Editing: LF, KM, GT, KCW, LJ, JEC, NACJ and ST.

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COMPETING INTERESTS

The authors declare no competing interests.

ETHICAL APPROVAL

This study was determined to be exempt by the Drexel University Institutional Review Board (Protocol #2303009775). All methods were performed in accordance with the relevant guidelines and regulations.

ADDITIONAL INFORMATION

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41370-026-00976-2>.

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